

ELEC4631 Solutions Tutorial 8

1. (a)

$$\begin{aligned}
 & (\alpha x_1 + (1 - \alpha)x_2)^2 - \alpha x_1^2 - (1 - \alpha)x_2^2 \\
 &= \alpha^2 x_1^2 + 2\alpha(1 - \alpha)x_1x_2 + (1 - \alpha)^2 x_2^2 - \alpha x_1^2 - (1 - \alpha)x_2^2, \\
 &= \alpha(\alpha - 1)x_1^2 + 2\alpha(1 - \alpha)x_1x_2 + (1 - \alpha)(1 - \alpha - 1)x_2^2, \\
 &= -\alpha(1 - \alpha)(x_1^2 - 2x_1x_2 + x_2^2), \\
 &= -\alpha(1 - \alpha)(x_1 - x_2)^2, \\
 &\leq 0,
 \end{aligned}$$

since $\alpha \geq 0$ and $1 - \alpha \geq 0$. This holds for any $x_1, x_2 \in \mathbb{R}$, $0 \leq \alpha \leq 1$.

(b) Since X is symmetric, one has the decomposition $X = WDW^T$, with D a diagonal matrix with diagonal elements $d_1, d_2, \dots, d_n$ (from top left to bottom right) and W is orthogonal, $W^{-1} = W^T$. Then $X^{-1} = WD^{-1}W^T$. Since $X > 0$, all the diagonal element of D are positive. Hence all the diagonal elements of D^{-1} are also positive since they are $1/d_1, 1/d_2, \dots, 1/d_n$ (from top left to bottom right). The same argument goes the other way around to show $X^{-1} > 0 \rightarrow X > 0$.

(c) Let W be any invertible $n \times n$ matrix. We first show that $X > 0 \Rightarrow W X W^T > 0$. Indeed, for any $x \in \mathbb{R}^n$, $z = W^T x \neq 0$ whenever $x \neq 0$ by the invertibility of W . Hence, $x^T W X W^T x = z^T X z > 0$ for all $x \neq 0$, meaning that $W X W^T$ is positive definite.

By the same argument, one can show the converse, $W X W^T > 0 \Rightarrow X > 0$. Take $x = W^{-T} z$ (here $W^{-T} = (W^{-1})^T = (W^T)^{-1}$). Since W is invertible (hence also W^{-T}), $z \neq 0$ whenever $x \neq 0$. Hence, $z^T X z = x^T W X W^T x > 0$, for any $x \neq 0$. Therefore, $X > 0$.

2. (a) $H = \{(x, y) \in \mathbb{R}^2 \mid x \geq -a, x \leq a, y - b - x \leq 0, y - b + x \leq 0, y + b - x \geq 0, y + b + x \geq 0\}$. Take any two points $(x_i, y_i) \in H$, $i = 1, 2$. Let $x_\alpha = \alpha x_1 + (1 - \alpha)x_2$ and $y_\alpha = \alpha y_1 + (1 - \alpha)y_2$ for $0 \leq \alpha \leq 1$. Then we can verify that (shown in the

tutorial):

$$x_\alpha \geq -a, \quad (1)$$

$$x_\alpha \leq a, \quad (2)$$

$$y_\alpha - b - x_\alpha \leq 0, \quad (3)$$

$$y_\alpha - b + x_\alpha \leq 0, \quad (4)$$

$$y_\alpha + b - x_\alpha \geq 0, \quad (5)$$

$$y_\alpha + b + x_\alpha \geq 0, \quad (6)$$

that holds for all $0 \leq \alpha \leq 1$. We conclude that $x_\alpha, y_\alpha \in H$ for all $0 \leq \alpha \leq 1$ for any two points (x_1, y_1) and (x_2, y_2) in H . Thus, H is a convex set.

- (b) $P = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq r, y - 3x \leq 0, y - x \geq 0\}$. Take any two points $(x_i, y_i) \in P, i = 1, 2$. As before, let $x_\alpha = \alpha x_1 + (1 - \alpha)x_2$ and $y_\alpha = \alpha y_1 + (1 - \alpha)y_2$ for $0 \leq \alpha \leq 1$. Then we can verify that (shown in the tutorial):

$$x_\alpha^2 + y_\alpha^2 \leq r, \quad (7)$$

$$y_\alpha - 3x_\alpha \leq 0, \quad (8)$$

$$y_\alpha - x_\alpha \geq 0, \quad (9)$$

that holds for all $0 \leq \alpha \leq 1$. We conclude that $x_\alpha, y_\alpha \in P$ for all $0 \leq \alpha \leq 1$ for any two points (x_1, y_1) and (x_2, y_2) in P . Thus, P is a convex set.

3. Suppose $C_1 \cap C_2 \neq \emptyset$ (i.e., not empty). Let $x_1, x_2 \in C_1 \cap C_2 \Rightarrow x_1, x_2 \in C_1$ and $x_1, x_2 \in C_2 \Rightarrow \alpha x_1 + (1 - \alpha)x_2 \in C_1$ and $\alpha x_1 + (1 - \alpha)x_2 \in C_2$ for all $0 \leq \alpha \leq 1$ (since C_1 and C_2 are individually convex). Therefore, $\alpha x_1 + (1 - \alpha)x_2 \in C_1 \cap C_2$ for all $0 \leq \alpha \leq 1$ for any $x_1, x_2 \in C_1 \cap C_2$ and we conclude that if $C_1 \cap C_2$ is not empty then it is a convex set.

Now, suppose that $C_1, C_2, \dots, C_k, \dots, C_n$ are convex sets for any n , and $C_1 \cap C_2 \cap \dots \cap C_n$ is not empty. We will show that $C_1 \cap C_2 \cap \dots \cap C_n$ is convex by mathematical induction. C_1 is a convex set, so suppose that $C_1 \cap C_2 \cap \dots \cap C_k$ is convex for $k \leq n - 1$. We will show that this implies that $C_1 \cap C_2 \cap \dots \cap C_k \cap C_{k+1}$ is also convex. Indeed, $C_1 \cap C_2 \cap \dots \cap C_k \cap C_{k+1} = (C_1 \cap C_2 \cap \dots \cap C_k) \cap C_{k+1}$. Since $C_1 \cap C_2 \cap \dots \cap C_k$ and C_{k+1} are convex and their intersection is non-empty, it follows that $C_1 \cap C_2 \cap \dots \cap C_k \cap C_{k+1}$ is convex (by what we had shown earlier). This statement holds true for k up to and including $n - 1$ by mathematical induction, and is also valid for any n .

$$4. F(x) = \begin{bmatrix} -3x_1 + x_2 & -x_1 + x_2 - 1 \\ -x_1 + x_2 - 1 & x_1 + x_2 \end{bmatrix} > 0.$$

$$(a) c_{F(x)}(\lambda) = \det(\lambda I - F(x)) = \lambda^2 + 2(x_1 - x_2)\lambda - 4x_1^2 - 2x_1 + 2x_2 - 1.$$

- (b) Since all eigenvalues of $F(x)$ must be real (because $F(x)$ is symmetric) the condition for $c_{F(x)}(\lambda)$ to have positive roots is $x_1 - x_2 < 0$ and $-4x_1^2 - 2x_1 + 2x_2 - 1 > 0$. This can be simplified to $x_1 - x_2 < 0$ and $x_2 > 2x_1^2 + x_1 + 1/2$.
- (c) The area in $\mathbb{R}^2$ defined by these two inequalities is not empty and can be found by sketching the region satisfying the two inequalities as shown in Fig. 1. The shaded region in the figure corresponds to $x_2 > x_1$ and $x_2 > 2x_1^2 + x_1 + 1/2$. So, this LMI is feasible.

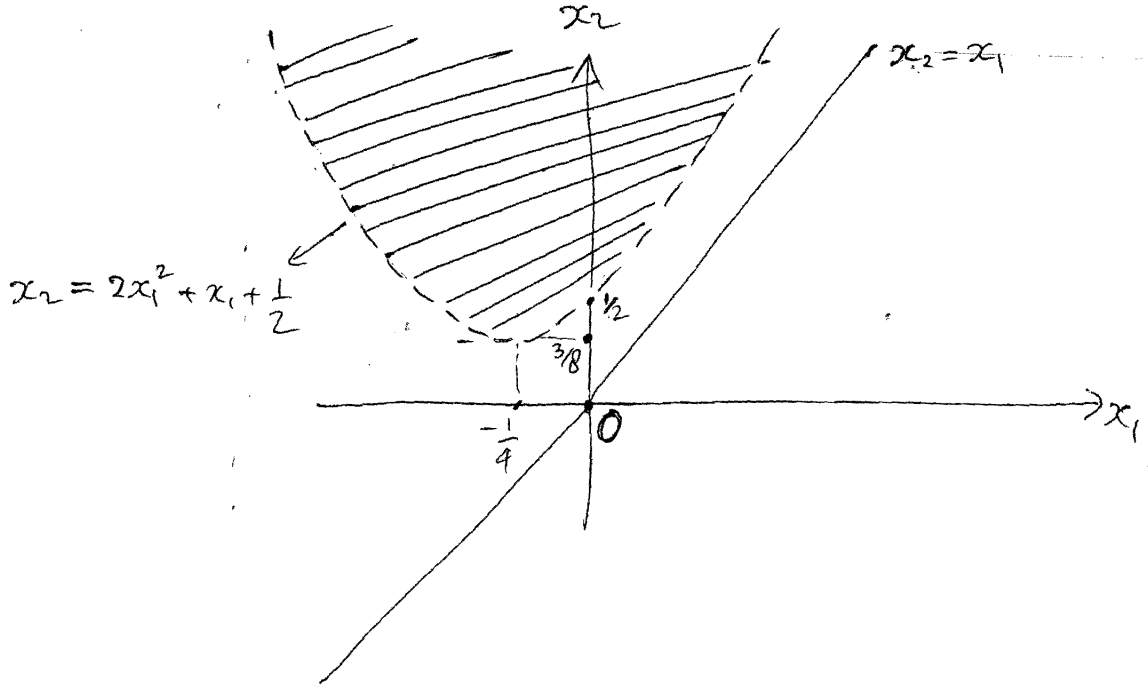


Figure 1: Shaded area shows the convex set of the solutions to the LMI of Problem 4(c). The LMI is feasible.

5. This can be done with two different methods yielding the same answer.

Method 1. $X > 4I$ is equal to the LMI

$$\begin{bmatrix} x_1 - 4 & x_2 \\ x_2 & x_3 - 4 \end{bmatrix} > 0.$$

To deal with the second LMI, expand X as (since it is symmetric):

$$X = x_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The left hand side of the second LMI becomes:

$$\begin{aligned}
& \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} - X \\
&= \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \left(x_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right) \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix}, \\
&= x_1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 & 2 \\ 2 & 6 \end{bmatrix} + x_3 \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix} - \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix}, \\
&= \begin{bmatrix} 4x_3 - x_1 & x_2 + 6x_3 \\ x_2 + 6x_3 & x_1 + 6x_2 + 8x_3 \end{bmatrix}.
\end{aligned}$$

The entire LMI in the vector variable $x = (x_1, x_2, x_3)$ is then:

$$\begin{bmatrix} x_1 - 4 & x_2 & 0 & 0 \\ x_2 & x_4 - 4 & 0 & 0 \\ 0 & 0 & x_1 - 4x_3 & -x_2 - 6x_3 \\ 0 & 0 & -x_2 - 6x_3 & -x_1 - 6x_2 - 8x_3 \end{bmatrix} > 0.$$

Method 2. The first LMI is the same as before. For the second LMI, do a direct mulitplication,

$$\begin{aligned}
& \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} - X \\
&= \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix}, \\
&= \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2x_2 & x_1 + 3x_2 \\ 2x_3 & x_2 + 3x_3 \end{bmatrix} - \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix}, \\
&= \begin{bmatrix} 4x_3 - x_1 & x_2 + 6x_3 \\ x_2 + 6x_3 & x_1 + 6x_2 + 8x_3 \end{bmatrix}.
\end{aligned}$$

The entire LMI in the vector variable $x = (x_1, x_2, x_3)$ is then the same as the one obtained via Method 1, as it should be.